

Project Report: Analyzing and Predicting Used Cars Prices in Canada

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1 Project Overview

This project uses the **Used cars listings for US & Canada** dataset to study how car prices relate to miles driven, car's make, province in which it is listed, year it was manufactured, and type of car (like SUV, Sedan, etc.). The report firsts analysis these features to show how well they contribute to the price of a car. And it compares different models with single or multiple features: miles vs miles, car age, make and province to determine whether simple or multiple features are enough to support useful prediction.

This question matters because someone looking for a first time used car would need to go through the process of what impacts a car's price and what would be considered a good deal. Using different model could help estimate and understand a car's value in the market. It not only tells where to buy for a better deal but it also gives insight on the factors to consider when looking for a first time car.

The main result is that the multi feature (miles, car age, body type, province and make) are highly informative. After keeping 5 key features along with 1 target variable and removing 68,977 incomplete rows and applying feature engineering techniques such as one-hot encoding and target encoding, multiple linear regression models were developed and evaluated using both a simple train-test split and full-data comparison. A simple linear regression model using only miles performed poorly, with a test R^2 of approximately 0.179, showing that miles alone is insufficient to explain price variation. In contrast, a multiple linear regression model incorporating miles, car age, make, body type, and province achieved a significantly im-

proved test R^2 of approximately 0.715 and a much lower test MSE 686.864. A gradient descent-based model trained on the same features produced nearly identical performance, confirming the stability of the optimization process and the reliability of the learned parameters. Over all, the predicted prices form the regression model with multiple features was further used to evaluate "good deals" in the dataset by comparing predicted vs listed price and identified the province which have statistically better deals.

2 Research Question

This report answers four specific questions:

1. Is mileage alone sufficient to predict used car prices, or are additional features required for accurate valuation?
2. Which car types (body types and makes) tend to offer the best value for first-time buyers in the Canadian used car market?
3. Which regression model provides the most accurate and reliable predictions for used car prices?
4. Which Canadian provinces show the highest frequency of undervalued used car listings based on model-predicted prices?

These questions are measurable because the dataset includes observed used-car listings with numeric variables such as miles, year (used to get car age), and price, as well as categorical features like make, body type, and province. These variables allow both regression modeling and comparison of predicted versus actual prices to evaluate model accuracy and identify undervalued vehicles.

Rows (raw / cleaned)	393603 / 324626
Columns in raw dataset	21
Time range	1981–2022
Missing values present?	Yes — 68977 rows removed

Table 1: Basic facts about the dataset used in this example.

3 Data Source and Description

3.1 Where the data came from

The dataset used in this project is the **Used cars listings for US & Canada** dataset. Each row represents one car listed on a Canadian car market either independent or from a dealership, with price, mileage, province location, make, year, and body type.

- **Dataset name:** Used cars listings for US & Canada
- **Link:** <https://www.kaggle.com/datasets/rupesthbraundal/marketcheck-automotive-data-us-canada>
- **What each row represents:** one unique car for sale
- **Key columns used:** price, miles, make, year, body_type and state
- **Licensing/usage note:** the dataset is released under an “Other (specified in description)” license on Kaggle, so it is permissible for analysis use but not for commercial or redistribution. Therefore, this project uses the dataset strictly for academic and educational purposes.

3.2 Basic dataset facts

After using the cleaned working dataset for visualization, for prediction I implemented one-hot encoding, target encoding and updated year columns to give appropriate weights to categorical data.

4 Data Cleaning and Feature Engineering

The raw dataset needed some work, as it had many missing values and a lot of categorical values.

- Removed 68977 incomplete rows (17.5% of the dataset) because at least one required field was missing. I chose dropping over imputation because key values like price were missing and other values were non-imputable.
- Checked for duplicate rows and found none.
- Verified that year was numeric and that all measurement columns were stored as numeric values.
- One-hot encoded **province** and **body_type** inside the modeling pipeline.
- Target encoded **make** by assigning mean values of each make.
- Updated **year** column to represent how old the car is related to newest car in the dataset.
- Make new columns to store predicted values from each Prediction Model.

These feature choices were based on the meaning of the variables in the used car dataset. Raw features like miles and year capture basic information about usage and age. Categorical features such as make, body type, and province were encoded so they could be used in the regression models. These transformations help the model better capture price differences without needing more complex nonlinear methods.

5 Exploratory Data Analysis and Visualization

The exploratory analysis focused on how the average price of a car is related to the five features in the dataset.

5.1 Plot 1: By Province

Figure 1 shows that the average car values are very different depending on the province of the country. Getting a car from say Northern Territories would be statistically more expensive than compared to in Prince Edward Island.

5.2 Plot 2: By Make

Figure 2 shows top and bottom 5 car brands with their average prices. The results highlight

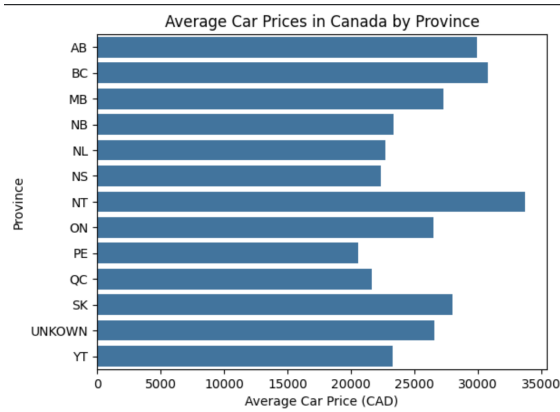


Figure 1: Prices of cars is cheaper in Prince Edwards Island than any other province.

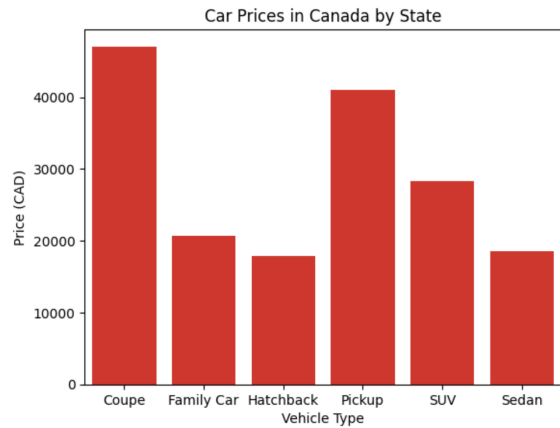


Figure 3: Getting a Hatchback or a sedan would be an affordable decision over a coupe or an SUV.

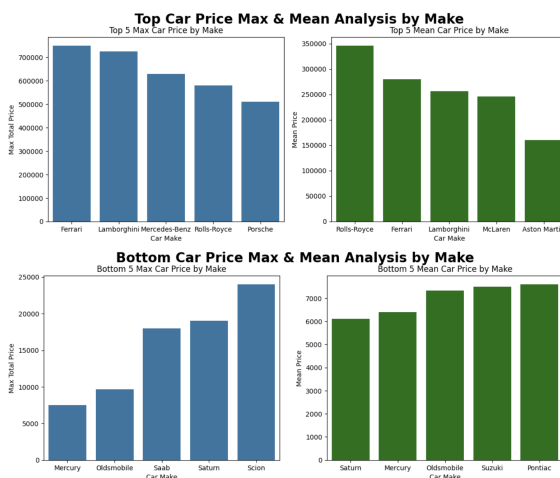


Figure 2: Brands like Ferrari and Rolls Royce are more expensive and Mercury and Saturn are cheaper.

significant variation in pricing across different makes, with values ranging from approximately \$7,000 to over \$7,000,000. On average, higher-end brands are priced around \$350,000, indicating strong brand influence on used car prices.

5.3 Plot 3: By Body Type

Figure 3 illustrates how prices vary across different body types. Coupes tend to be the most expensive, as many high-end luxury and sports vehicles are designed in this style. In contrast, hatchbacks, sedans, and family-oriented vehicles such as vans are generally more affordable, making them more suitable options for first-time buyers and everyday commuting.

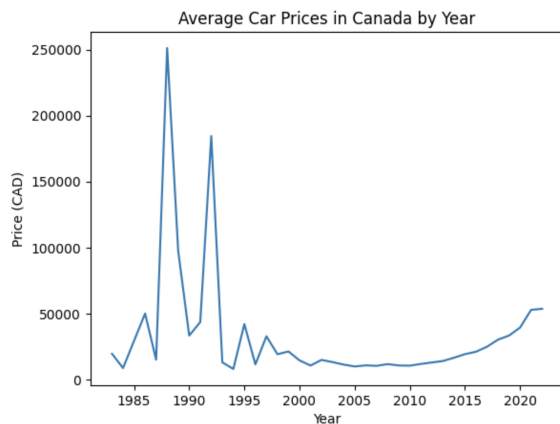


Figure 4: Cars in between 1985 and 1995 have higher price, possibly due to rarity in the market.

5.4 Plot 4: By Year

Figure 4 shows how car prices vary with model year. In general, older vehicles tend to be less expensive than newer ones due to depreciation. However, some older model years still have high prices, which can be attributed to rarity, collectible value, or limited availability of certain vehicles.

5.5 Plot 5: By Mileage

Upon careful inspection Figure 5 shows a negative relationship between the two variables, where prices generally decrease as mileage increases. Most data points are clustered around approximately 60,000 miles and \$40,000, indicating that a large portion of the dataset consists of mid-range vehicles. While the overall trend reflects depreciation with higher mileage, some variation remains due to differences in make, model, and vehicle condition.

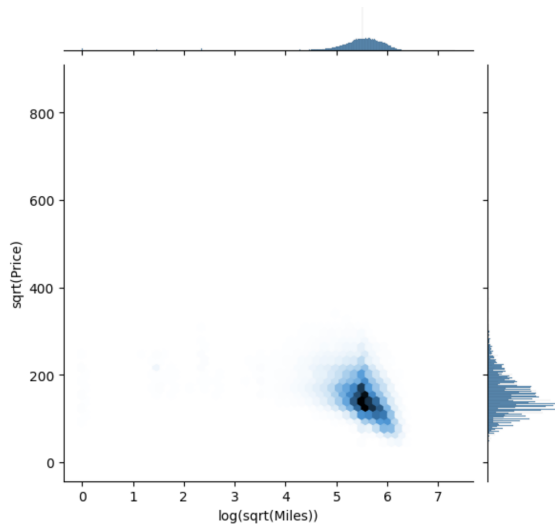


Figure 5: $(e^{5.5})^2 \approx 60,000$ miles and the price is around $200^2 = \$40,000$.

6 Data Analysis

6.1 Analysis Plan

I made three regression models to predict used car prices. A simple regression model (SLR1) with just miles as a feature to serve as a baseline. A linear regression model (SLR2) with multiple features miles, car age, make, body type, and province, with categorical variables encoded using one-hot and target encoding. Finally, a gradient descent-based linear regression model was implemented using the same set of features to validate parameter optimization.

I initially trained the models with all datasets to find the R^2 and MSE value different between the 3 models and then trained each model in a 80-20 split to compare the underfitting or overfitting of the model.

6.2 Baseline

For selecting a suitable model, the baseline used only miles to predict price. It provided a reference point for evaluating more complex models but underfitted the data, indicating that additional features are necessary for accurate price prediction.

6.3 Main Results

Table 2 shows a substantial improvement in performance when moving from the baseline model to more advanced linear regression models. The baseline model (SLR1) has a high er-

Model	Overall MSE	Overall R^2
SLR1	1987.4	0.175
SLR2	686.8	0.715
Gradient Descent)	686.8	0.715

Table 2: Multiple Linear Regression and Gradient Descent performed better than Simple Linear Regression, with the least amount of Loss and showing 71.5% of variance.

Model	Train MSE	Test MSE
SLR1	1987.7	1986.5
SLR2	685.0	694.0
Gradient Descent	686.1	695.4

Table 3: In the 80-20 split Linear Regression with all the features showed the least amount of error, though Gradient Descent is almost same.

ror, with an RMSE of approximately \$1,987.4, while the best-performing model reduces this error to about \$686.8. Simple linear regression is weak here although, when using multiple features (SLR2), explains a lot of variance in the data as shown by the R^2 values. Using just mileage explains only 17.5% of the variance in the data. While using all the other features increase the variance to 71.5% demonstration importance of the other features. Using Gradient Descent model gives almost identical results as Linear Regression with multiple feature, which is a strong indicator that the models are performing well.

Table 3 shows that when using an 80-20 train-test split, the MSE values remain very similar to those obtained using the full dataset. This consistency indicates that the models generalize well and do not exhibit signs of overfitting. For both SLR2 and the gradient descent model, the training and testing loss values are very close, further confirming the stability and reliability of the learned models.

Table 4 demonstrates that, even with the train-test split, the test performance remains very consistent across models, indicating strong generalization. This supports the use of multiple features for predicting used car prices. Although SLR2 and the gradient descent model achieve nearly identical R^2 values, the multiple linear regression model is preferred due to its slightly better performance

From Figure 6 the most amount of good deals are found in Ontario. The good deals are com-

Model	Train R^2	Test R^2
SLR1	0.174	0.179
SLR2	0.715	0.713
Gradient Descent	0.715	0.712

Table 4: In the test split again SLR2 explain around 71.3% of variance in the data, Gradient is the close second with 71.2%.

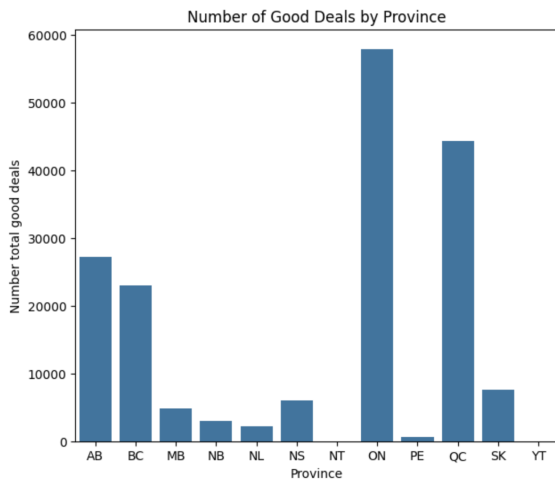


Figure 6: Ontario has the most number of good deals.

puted by comparing the predicted price with listed price and counting all the good deals in a province.

6.4 Interpretation

The results show that considering just miles when buying for a used car is not enough, people should consider other factors like time car was used, make, body type, and location-when buying. The models with considered these factors showed around 71% of variation in car prices. The analysis also suggested that car types, likes hatchback and sedans, tend to be more affordable and buying cars from certain province may get a better deal.

One surprising observation was that the gradient descent model did not outperform linear regression with multiple features, despite being more computational. This indicate that the relationship of the data is more linear.

Few limitation in this analysis are that not all factors were considered such as accident history and vehicle condition. The data could be distributed unevenly like provinces with high populations would have more listing making the data skewed.

7 Real-World Application

A realistic stakeholder is **someone who is looking for their first car** who needs guideline on what listing is a good listing for a car price.

Based on this example, they could make the following decisions:

- Use a linear model as a quick estimate to find the car's estimate value.
- Use insights from the data visualization to choose a car by made and body type.
- Find where to get a car from, like Ontario, Quebec or Alberta.

My recommendation would be to use the linear-regression model for approximate car price estimation. Confidence in that recommendation is **moderate**: variation of the car price is 71% in this dataset, but using the model does give a really good idea on what actual price should be.

8 Report Quality and Reproducibility

8.1 How to run the code

- **Code** **location:** <https://github.com/Dharmrajsinh13/Analyzing-and-predicting-used-cars-in-canada>.
- **Main** **notebook:** `Analysing_and_Predicting_Used_Canadian_Car_Prices.ipynb`
- **Dataset:** <https://www.kaggle.com/datasets/rupeshraund/automotive-data-us-canada>
- **Suggested steps:**
 - Clone or download the project repository from GitHub.
 - Install the required Python packages: `pip install -r requirements.txt`
 - Download the *Used Cars Listings for US & Canada* dataset from Kaggle.
 - Upload the dataset to the Colab run-time and update the dataset path in the notebook if necessary. (If you get a warning about the run-time file upload, ignore it and click ok.)

- Run all the steps one after another or click run all button.
- Run the modeling section to reproduce Tables 2, 3 and 4.

8.2 Project structure

- README.md
- requirements.txt
- notebooks/
- report/
- figures/

A strong report is reproducible as well as readable. Another student should be able to identify the dataset, repeat the cleaning choices, rerun the train/test split, and regenerate the plots and tables without guessing what happened.

9 Conclusion

Conclusion: This project shows that there are several factors one should consider when buying a used car. Using a multiple linear regression with all those factor gives a better results as the model will show a better variance to car price. And buying a car at certain locations is wise.

10 AI Disclosure

- **Tools used:** ChatGPT based on the GPT-5.3 architecture for wording and editing support.
- **What it was used for:** revising paragraph clarity, editing graphs to visualize, editing and debugging support, finding cause of errors, and checking grammar.

References

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